



Predicting Mental Health Trends Using Reddit Discussions

Welcome to NeuroNexus's presentation on predicting mental health trends using Reddit discussions. We are Group 11, and our team consists of Aman Ostwal, Sai Pokuri, Sanjoli Sogani, and Darshit Rai. This presentation summarizes our work for DSCI 592 – Capstone Phase II, building upon our previous work to analyze mental health discussions on Reddit.

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Project Overview: NeuroNexus



Reddit Analysis

Analyze Reddit discussions from mental health subreddits to understand community concerns.



NLP + ML

Predict disorder type and severity using Natural Language Processing and Machine Learning techniques.



Early Detection

Enable early detection of mental health issues through scalable data analysis.

Our project focuses on analyzing Reddit discussions from mental health subreddits to predict disorder type and severity. By leveraging NLP and ML, we aim to provide early detection, personalized insights, and a scalable pipeline for mental health analysis. This project seeks to transform online discussions into actionable insights for mental health support.



[illegible]

Collected over 17,000 posts using the PRAW API to gather a substantial dataset.

Preprocessed text data by cleaning and lemmatizing to prepare it for analysis.

Conducted EDA using word clouds, sentiment analysis, and posting patterns to understand data characteristics.

Simulated severity scoring via keyword rules to create initial labels for model training.

Made with **Gamma**

DSCI 592 Objectives: Modeling and Optimization

ML Models

Build machine learning models for severity and disorder classification to automate analysis.

Mental Health Score

Predict mental health scores based on text data to provide quantitative assessments.

Data Pipeline

Optimize and modularize the entire data pipeline for efficiency and scalability.

Ethical Use

Ensure clinical interpretability and ethical use of the models to maintain responsible AI.

In DSCI 592, our objectives are to build ML models for severity and disorder classification, predict mental health scores, optimize the data pipeline, and ensure clinical interpretability and ethical use. These objectives will enable us to create a robust and responsible system for mental health analysis.

Feature Engineering & Pipeline: Data Preparation

1

Tokenization

Break down text into individual tokens for analysis.

2

TF-IDF + n-grams

Use TF-IDF and n-grams (unigrams, bigrams, trigrams) to capture text importance and context.

3

Medication Mentions

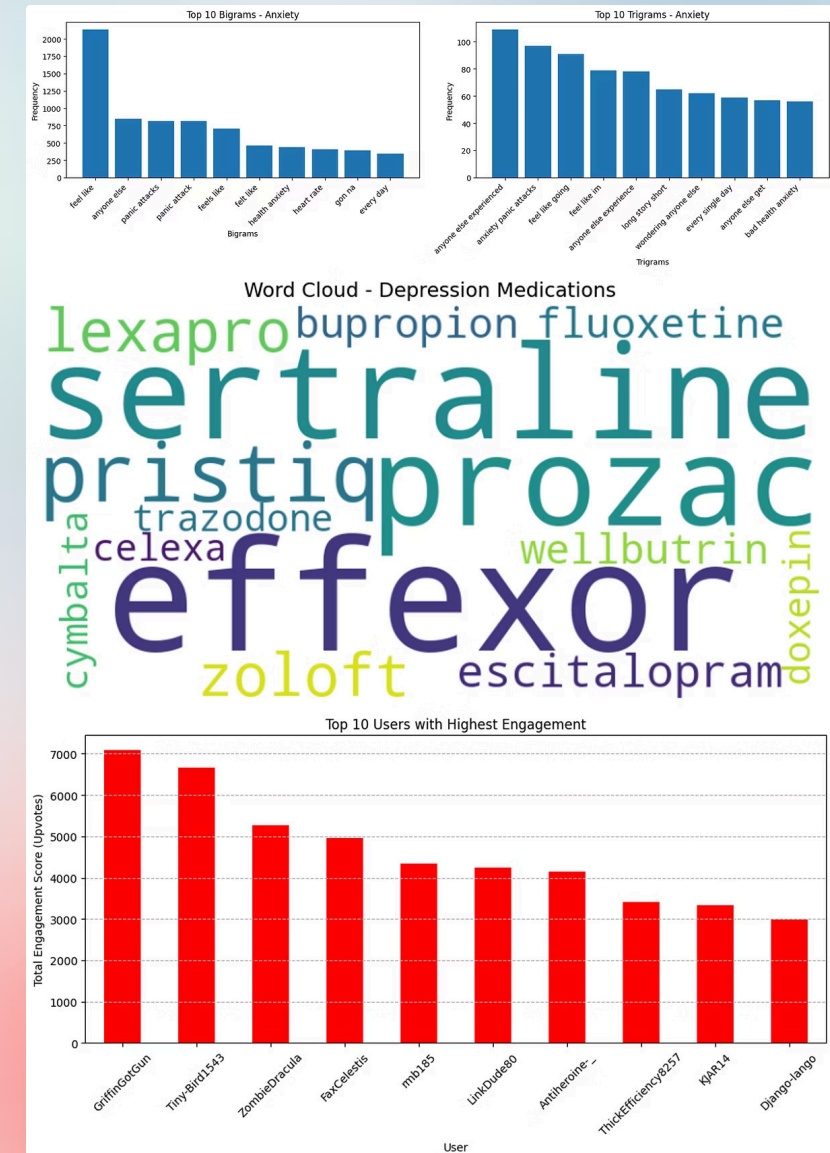
Include medication mentions to identify potential treatment contexts.

4

Engagement Metrics

Incorporate engagement metrics to understand user interaction patterns.

Our feature engineering process involves tokenization, TF-IDF with n-grams, medication mentions, and engagement metrics. By merging these input features, we aim to create a comprehensive dataset for effective modeling and analysis of mental health discussions.



Severity & Score Prediction: Regression Modeling



For severity and score prediction, we employ rule-based scoring using keywords, modifiers, and medications. We then train a Random Forest Regressor on simulated labels and adjust the scores to clinical thresholds for practical interpretation. This approach allows us to provide meaningful severity assessments.

Disorder Classification: Identifying Mental Health Conditions

1

TF-IDF + Keywords

Use TF-IDF and keyword-labeled data from subreddit content.

2

Random Forest

Apply a Random Forest Classifier to predict disorder type.

3

Evaluation

Evaluate using accuracy, F1-score, precision, and recall.

In disorder classification, we use TF-IDF and keyword-labeled data from subreddit content. A Random Forest Classifier is then applied to predict disorder type, and the results are evaluated using accuracy, F1-score, precision, and recall. This process helps us identify specific mental health conditions discussed on Reddit.

Learning Approaches & Evaluation: Model Selection

Supervised Learning

- Random Forest
- XGBoost

Use supervised learning algorithms for predictive modeling.

Unsupervised Learning

- LDA + Clustering

Apply unsupervised learning for theme discovery.

Evaluation Metrics

- RMSE, MAE (regression)
- F1, Accuracy (classification)

Evaluate models using appropriate metrics.

We explore both supervised and unsupervised learning approaches, including Random Forest, XGBoost, LDA, and clustering. Models are evaluated using RMSE and MAE for regression, and F1-score and accuracy for classification. Hyperparameter tuning is performed using Grid/Random Search to optimize model performance.

Challenges & Expert Support: Overcoming Limitations



Data Limitation

Reddit data is limited to recent posts via PRAW.



Data Imbalance

Data imbalance across disorders and severity levels.



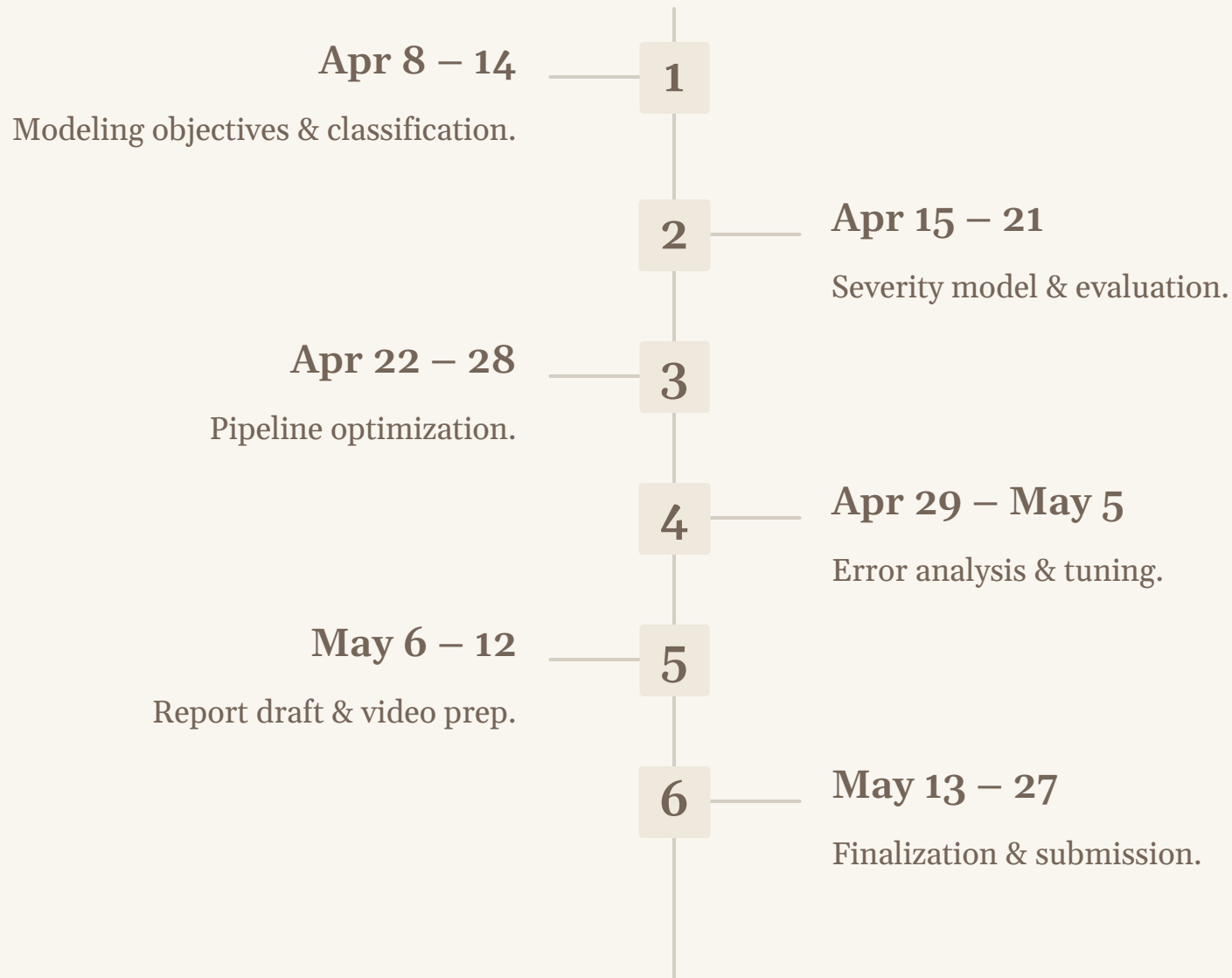
Avoid Leakage

Prevent leakage between rule-based and learned features.

We face challenges such as Reddit data limitations, data imbalance, and the need to avoid leakage between rule-based and learned features. Expert support from Ashleigh Zoerb & Nelca Ece Yilmaz guides clinical alignment, ensuring the project's relevance and ethical considerations are addressed.



Timeline & Next Steps: Project Completion



Our timeline includes modeling objectives and classification, severity model and evaluation, pipeline optimization, error analysis and tuning, report drafting and video preparation, and finalization and submission. We are committed to completing the project on time and delivering valuable insights into mental health trends using Reddit discussions.